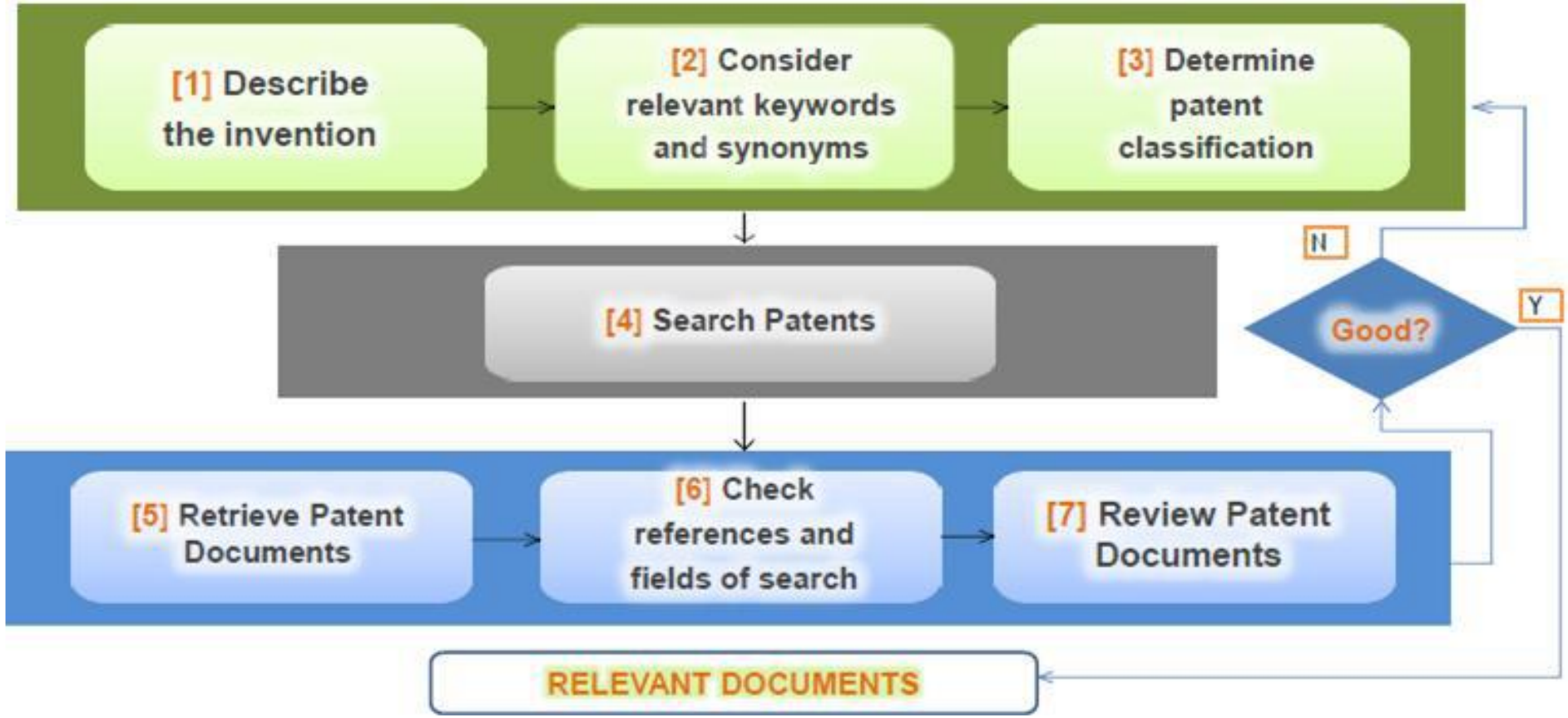
# Patent Search Flowchart



University of Science and Technology of Southern Philippines

Technology Promotions and Commercialization Office

# KEYWORD SEARCHING

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# Learning Objective

- Generate text query for keyword searching

# KEYWORD SEARCHING

- One of the basic strategies
- Keywords are taken from different parts of the document
- **Languages:** only EN documents retrieved with EN keywords
- Iterative kind of search until you exhaust the search

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
2. Use Problem-Solution Approach
3. Brainstorm synonyms, word variations or equivalents
4. Apply “ ” for phrases (more than one-word terms)
5. Apply word-stemming or truncation
6. Regroup terms related to a concept using Boolean operator
7. Use parentheses to organize search string and apply nesting

# STEPS IN CONDUCTING KEYWORD SEARCH

1. **Build out concepts and terms of technology**
2. Use Problem-Solution Approach
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# Step 1 - Build out concepts and terms of technology

## Sample Invention:

**A patch for administering vaccines that comprises microscopic projections made of silicon produced using deep reactive ion etching, with a length adapted to reach Langerhans cells, for improving immune responses through delivery and eliminating the cold chain.**

# Step 1 - Build out concepts and terms of technology

## Sample Invention:

**A patch for administering vaccines that comprises microscopic projections made of silicon produced using deep reactive ion etching, with a length adapted to reach Langerhans cells, for improving immune responses through delivery and eliminating the cold chain.**

### How to build out concepts and terms?

-terms on the problem does the invention solve
-terms on the structure of the invention
-terms on the function of the invention
-avoid broad and generic terms such as “device”, “process”, “system”, etc. Such terms will tend to bloat the search results.

# Step 1 - Build out concepts and terms of technology

## Sample Invention:

A patch for administering vaccines that comprises microscopic projections made of silicon produced using deep reactive ion etching, with a length adapted to reach Langerhans cells, for improving immune responses through delivery and eliminating the cold chain.

# Step 1 - Build out concepts and terms of technology

**Concepts/ terms:**

- immune response
- cold chain
- patch
- microscopic projection
- silicon
- deep reactive ion etching
- Langerhans cell
- vaccine

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
2. **Use Problem-Solution Approach**
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# Step 2 - Use Problem-Solution Approach

## Identify the following terms:

1. problem
2. structure
3. function

# Step 2 - Use Problem-Solution Approach

## Invention Diagram (Problem-Solution Approach):

| PROBLEM | SOLUTION | |
|---|---|---|
| | What it is: (structure) | What it does: (function) |
| | | |

# Step 2 - Use Problem-Solution Approach

**Concepts/ Terms from Step 1:**

- immune response
- cold chain
- patch
- microscopic projection
- silicon
- deep reactive ion etching
- Langerhans cell
- vaccine

# Step 2 - Use Problem-Solution Approach

## Invention Diagram:

| PROBLEM | SOLUTION | |
|---|---|---|
| | **What it is: (structure)** | **What it does: (function)** |
| immune response<br>cold chain | patch<br>microscopic projection<br>silicon<br>deep reactive ion etching<br>langerhans cell | vaccine |

# Step 2 - Use Problem-Solution Approach

## Invention Diagram:

| PROBLEM | SOLUTION | |
|---|---|---|
| | **What it is: (structure)** | **What it does: (function)** |
| immune response<br>cold chain | patch<br>microscopic projection<br>silicon<br>deep reactive ion etching<br>langerhans cell | vaccine |

# Step 2 - Use Problem-Solution Approach

## Solution (Structure, Function) Concepts/ Terms:

- vaccine
- patch
- microscopic projection
- silicon
- deep reactive ion etching
- langerhans cell

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
2. Use Problem-Solution Approach
3. **Brainstorm synonyms, word variations or equivalents**
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# Step 3 - Brainstorm synonyms, word variations or equivalents

- ✓ synonyms
- ✓ industry jargon
- ✓ legalese
- ✓ patentese

# Step 3 - Brainstorm synonyms, word variations or equivalents

- Search for **synonyms**

  e.g. Using specialized dictionaries

  **truck vs. lorry**

  **backpack vs. rucksack**

  **elevator vs. lift**

- Search for **word variations/ alternative spelling**

  e.g. **US plow** vs. **GB plough**

  **US color** vs. **GB colour**

# Step 3 - Brainstorm synonyms, word variations or equivalents

## Industry Jargons

- industry-specific terminology
- usage is limited to a specific industry

# Step 3 - Brainstorm synonyms, word variations or equivalents

**Legalese**

- the formal and technical language of legal documents

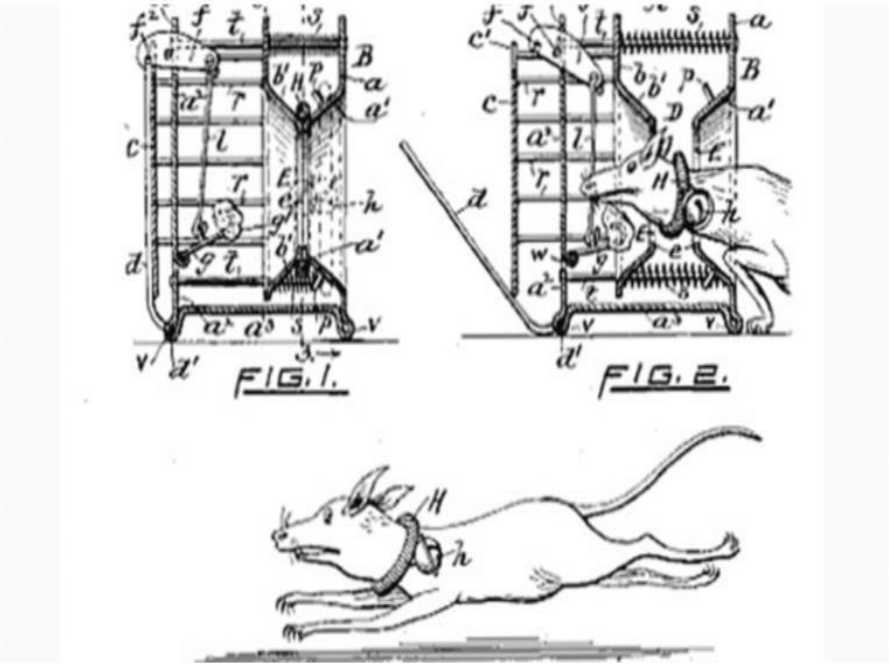
**Patentese**

- the legal jargon used in patents

# Step 3 - Brainstorm synonyms, word variations or equivalents

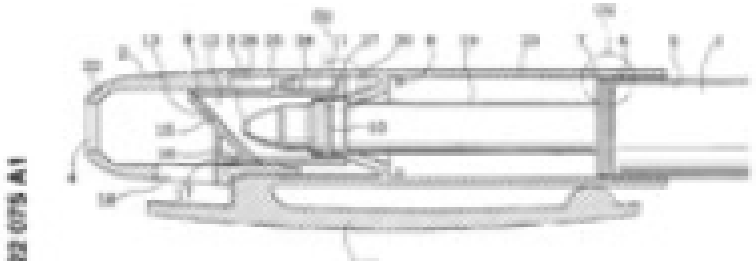
**Example:**

A "**mouse trap**" can also be a "**rodent extermination device**"



# Step 3 - Brainstorm synonyms, word variations or equivalents

pen



writing instrument

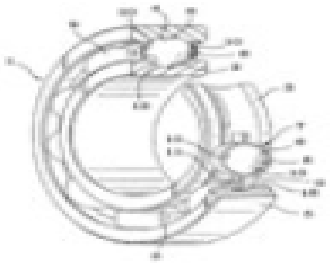
# Step 3 - Brainstorm synonyms, word variations or equivalents

toy ball

spherical object
with floppy filaments
to promote sure capture

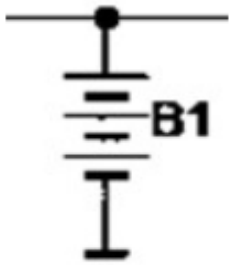
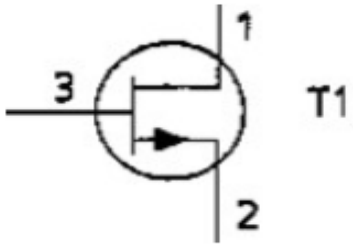
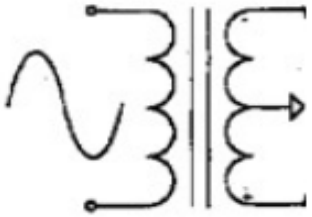
# Step 3 - Brainstorm synonyms, word variations or equivalents

ball bearing



a plurality of balls

# Step 3 - Brainstorm synonyms, word variations or equivalents

| | | |
|---|---|---|
| electrical power source for electronic circuits | = battery |   |
| semiconductor switching device with a control electrode | = transistor |   |
| galvanically isolated electrical coupling means | = transformer |  |

# Step 3 - Brainstorm synonyms, word variations or equivalents

Patents that use very different phrases in place of **"camera":**

- **Photographing optical apparatus (US5727239)**
- **Image pickup device** with facial region detector and method of synthesizing image including facial region **(US7440013)**
- **Digital photographing device** with separate optical distortion correction for dynamic images and still images **(US7265787)**
- **Image sensing apparatus (US7671896)**
- **Recording apparatus** and method, reproducing apparatus and method, and program **(US8520086)**
- **Image capturing system**, image capturing device, and image capturing method **(US8369701)**
- Eye examining instrument with **photo-detecting system (US4257687)**

# Step 3 - Brainstorm synonyms, word variations or equivalents

**LANGUAGE**

Not all patent documents are written in English

**ALTERNATIVE SPELLINGS**

color vs. colour; plow vs. plough; center vs. centre

**WORD VARIATIONS**

truck vs. lorry; backpack vs. rucksack;
diaper vs. nappy; elevator vs. lift

# Step 3 - Brainstorm synonyms, word variations or equivalents

- vaccine
- patch
- microscopic projections
- silicon
- deep reactive ion etching
- langerhans cell

# Step 3 - Brainstorm synonyms, word variations or equivalents

- vaccine: inoculation, antigen, antibody generator
- patch
- microscopic projections: microprojections, nanoprojections, microscopic needles, microneedles, nanoneedles
- silicon
- deep reactive ion etching: DRIE
- langerhans cell: dendritic cell, antigen-presenting immune cell

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
2. Use Problem-Solution Approach
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4. **Apply “ ” for phrases (more than one word terms)**
5. Apply word-stemming or truncation
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# Step 4 - Apply “ ” for phrases (more than one-word terms)

- Use quotation marks “ ” when searching for a phrase

Examples:

“solar panel”

“electronic cigarette”

# Step 4 - Apply “ ” for phrases (more than one-word terms)

- vaccine, inoculation, antigen, antibody generator
- patch
- microscopic projections, microprojections, nanoprojections, microscopic needles, microneedles, nanoneedles
- silicon
- deep reactive ion etching, DRIE
- langerhans cell, dendritic cell, antigen-presenting immune cell

# Step 4 - Apply “ ” for phrases (more than one-word terms)

- vaccine, inoculation, antigen, “antibody generator”
- patch
- “microscopic projections”, microprojections, nanoprojections, “microscopic needles”, microneedles, nanoneedles
- silicon
- “deep reactive ion etching”, DRIE
- “langerhans cell”, “dendritic cell”, “antigen-presenting immune cell”

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
2. Use Problem-Solution Approach
3. Brainstorm synonyms, word variations or equivalents
4. Apply “ ” for phrases (more than one-word terms)
5. **Apply word-stemming or truncation**
6. Regroup terms related to a concept using Boolean operator
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# Step 5 - Apply word-stemming or truncation

- **asterisk (*) - used as a wildcard character that represents any number of characters.**

  - attach* - attach, attached, attaching, attachment, attachable
  - *attach - reattach (re-attach)
  - *attach* - reattaching (re-attaching), reattached (re-attached)

# Step 5 - Apply word-stemming or truncation

- **question mark (?) - used to represent a single character or a specific position in a word.**
  - speciali?ation – specialisation, specialization
  - commerciali?ation – commercialisation, commercialization

# Step 5 - Apply word-stemming or truncation

- vaccine, inoculation, antigen, “antibody generator”
- patch
- “microscopic projections”, microprojections, nanoprojections, “microscopic needles”, microneedles, nanoneedles
- silicon
- “deep reactive ion etching”, DRIE
- “langerhans cell”, “dendritic cell”, “antigen-presenting immune cell”

# Step 5 - Apply word-stemming or truncation

- vaccin*, inoculat*, antigen*, "antibody generat*"
- patch*
- "micro* projection*", microprojection*, nanoprojection*, "micro* needle*", microneedle*, nanoneedle*
- silicon
- "deep reactive ion etching", DRIE
- "langerhans cell*", "dendritic cell*", "antigen-presenting immune cell*"

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
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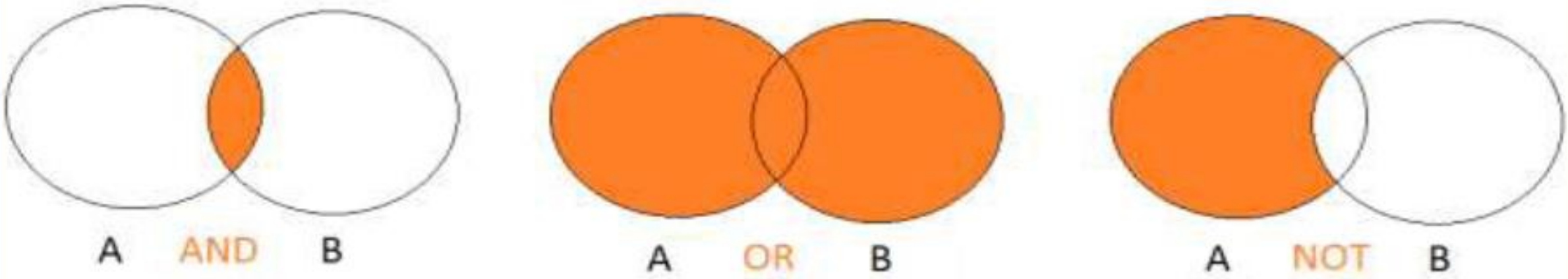
4. Apply “ ” for phrases (more than one-word terms)
5. Apply word-stemming or truncation
6. **Regroup terms related to a concept using Boolean operator**
7. Use parentheses to organize search string and apply nesting

# Step 6 - Regroup terms related to a concept using Boolean operator

## BOOLEAN OPERATORS: AND, OR, NOT



# Step 6 - Regroup terms related to a concept using Boolean operator

## BOOLEAN OPERATORS: AND, OR, NOT

- **tennis AND racket:** all docs having both the word tennis and racket
- **tennis OR racket:** all docs having either the word tennis or the word racket
- **tennis NOT racket:** all docs having the word tennis but not the word racket

# Step 6 - Regroup terms related to a concept using Boolean operator

- vaccin*, inoculat*, antigen*, "antibody generat*"
- patch*
- "micro* projection*", microprojection*, nanoprojection*, "micro* needle*", microneedle*, nanoneedle*
- silicon
- "deep reactive ion etching", DRIE
- "langerhans cell*", "dendritic cell*", "antigen-presenting immune cell*"

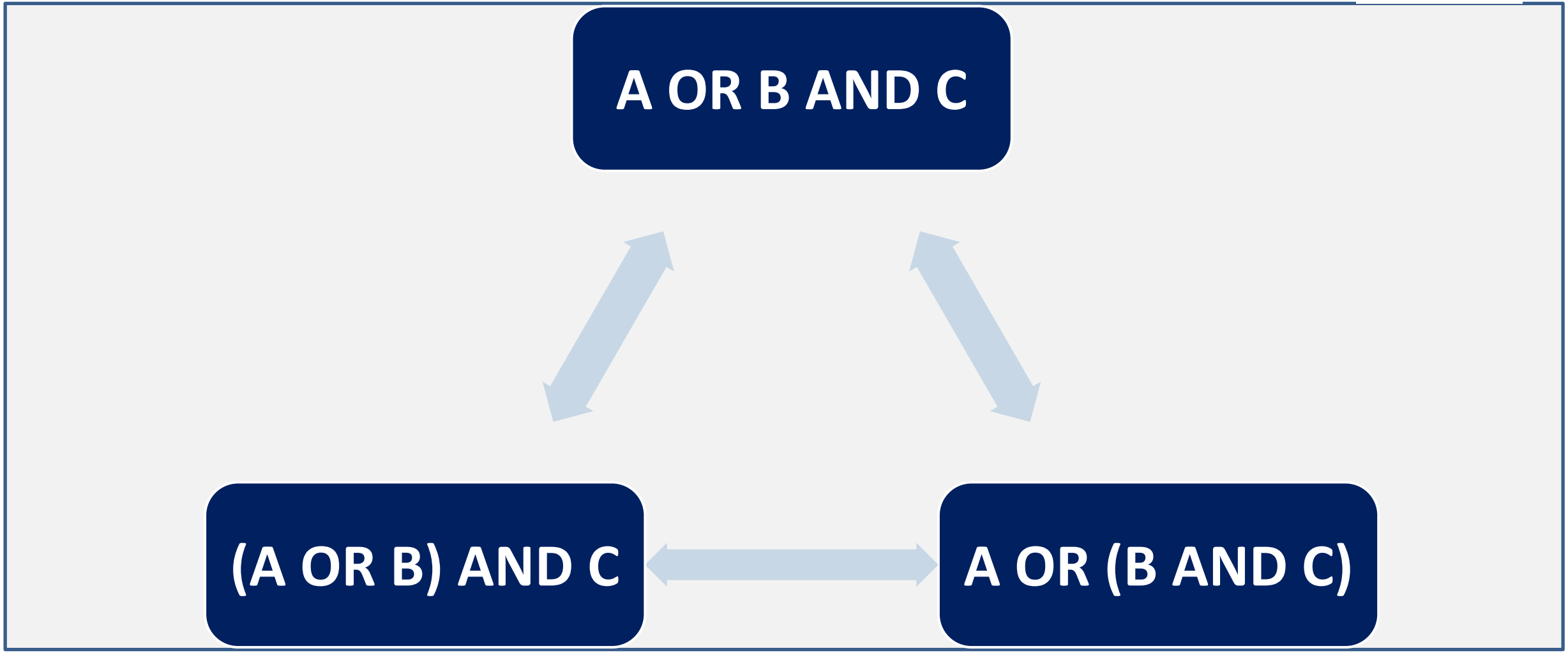
# Step 6 - Regroup terms related to a concept using Boolean operator

- vaccin* **OR** inoculat* **OR** antigen* **OR** "antibody generat*"
- patch*
- "micro* projection*" **OR** microprojection* **OR** nanoprojection* **OR** "micro* needle*" **OR** microneedle* **OR** nanoneedle*
- silicon
- "deep reactive ion etching" **OR** DRIE
- "langerhans cell*" **OR** "dendritic cell*" **OR** "antigen-presenting immune cell*"

# STEPS IN CONDUCTING KEYWORD SEARCH

1. Build out concepts and terms of technology
2. Use Problem-Solution Approach
3. Brainstorm synonyms, word variations or equivalents
4. Apply “ ” for phrases (more than one-word terms)
5. Apply word-stemming or truncation
6. Regroup terms related to a concept using Boolean operator
7. **Use parentheses to organize search string and apply nesting**

# Step 7 - Use parentheses to organize search string and apply nesting



# Step 7 - Use parentheses to organize search string and apply nesting

- **Used when keywords are nested to clarify the <u>order</u> of search query elements**

- **For example, you want to find patent documents on either red seaweeds or brown seaweed**

❑seaweed AND red OR brown → possible that a search database may confuse the search as:

❑(seaweed AND red) OR brown

✓seaweed AND (red OR brown)

# Step 7 - Use parentheses to organize search string and apply nesting

- **Wrong format**
  - Dog OR cat OR pet AND cage OR kennel OR housing

- **Using proper brackets**
  - (Dog OR cat OR pet ) AND (cage OR kennel OR housing)

# Step 7 - Use parentheses to organize search string and apply nesting

- vaccin* **OR** inoculat* **OR** antigen* **OR** "antibody generat*"
- patch*
- "micro* projection*" **OR** microprojection* **OR** nanoprojection* **OR** "micro* needle*" **OR** microneedle* **OR** nanoneedle*
- silicon
- "deep reactive ion etching" **OR** DRIE
- "langerhans cell*" **OR** "dendritic cell*" **OR** "antigen-presenting immune cell*"

# Step 7 - Use parentheses to organize search string and apply nesting

- (vaccin* OR inoculat* OR antigen* OR "antibody generat*")
- patch*
- ("micro* projection*" OR microprojection* OR nanoprojection* OR "micro* needle*" OR microneedle* OR nanoneedle* )
- silicon
- ("deep reactive ion etching" OR DRIE )
- ("langerhans cell*" OR "dendritic cell*" OR "antigen-presenting immune cell*")

# Step 7 - Use parentheses to organize search string and apply nesting

- (vaccin* OR inoculat* OR antigen* OR "antibody generat*") AND patch* AND ("micro* projection*" OR microprojection* OR nanoprojection* OR "micro* needle*" OR microneedle* OR nanoneedle* ) AND silicon AND ("deep reactive ion etching" OR DRIE ) AND ("langerhans cell*" OR "dendritic cell*" OR "antigen-presenting immune cell*")

# Final Text Query

- (vaccin* OR inoculat* OR antigen* OR "antibody generat*") AND patch* AND ("micro* projection*" OR microprojection* OR nanoprojection* OR "micro* needle*" OR microneedle* OR nanoneedle*) AND silicon AND ("deep reactive ion etching" OR DRIE) AND ("langerhans cell*" OR "dendritic cell*" OR "antigen-presenting immune cell*")

# PATENT DATABASES

| Database | Maintaining Office and Address |
|---|---|
| Espacenet | European Patent Office (EPO) |
| | https://worldwide.espacenet.com |
| Patentscope | World Intellectual Property Office (WIPO) |
| | https://www.wipo.int/patentscope/en/ |
| USPTO | US Patent and Trademark Office (USPTO) |
| | https://www.uspto.gov/patents-application-process/search-patents |
| IPDL | Japan's National Center for Industrial Property Information and Training (INPIT) |
| | https://www.j-platpat.inpit.go.jp |
| KIPRIS | Korean Institute of Patent Information |
| | http://eng.kipris.or.kr/enghome/main.jsp |

International Patent Classification | IPC Publication

wipo.int/classifications/ipc/ipcpub/?notion=search&version=20210101&symbol=none&menulang=en&lang=en&viewmode=p&fipc...

WIPO IP PORTAL | MENU | *IPC Publication* | Covid-19 Update | HELP | ENGLISH | WIPO

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2021.01 Version

English version

French version

Advanced Search

Terms

Cross-references

STATS

IPCCAT

**Terms search:**

Stemming

IPCPUB v8.5
**Last modified:** 2021.06.14

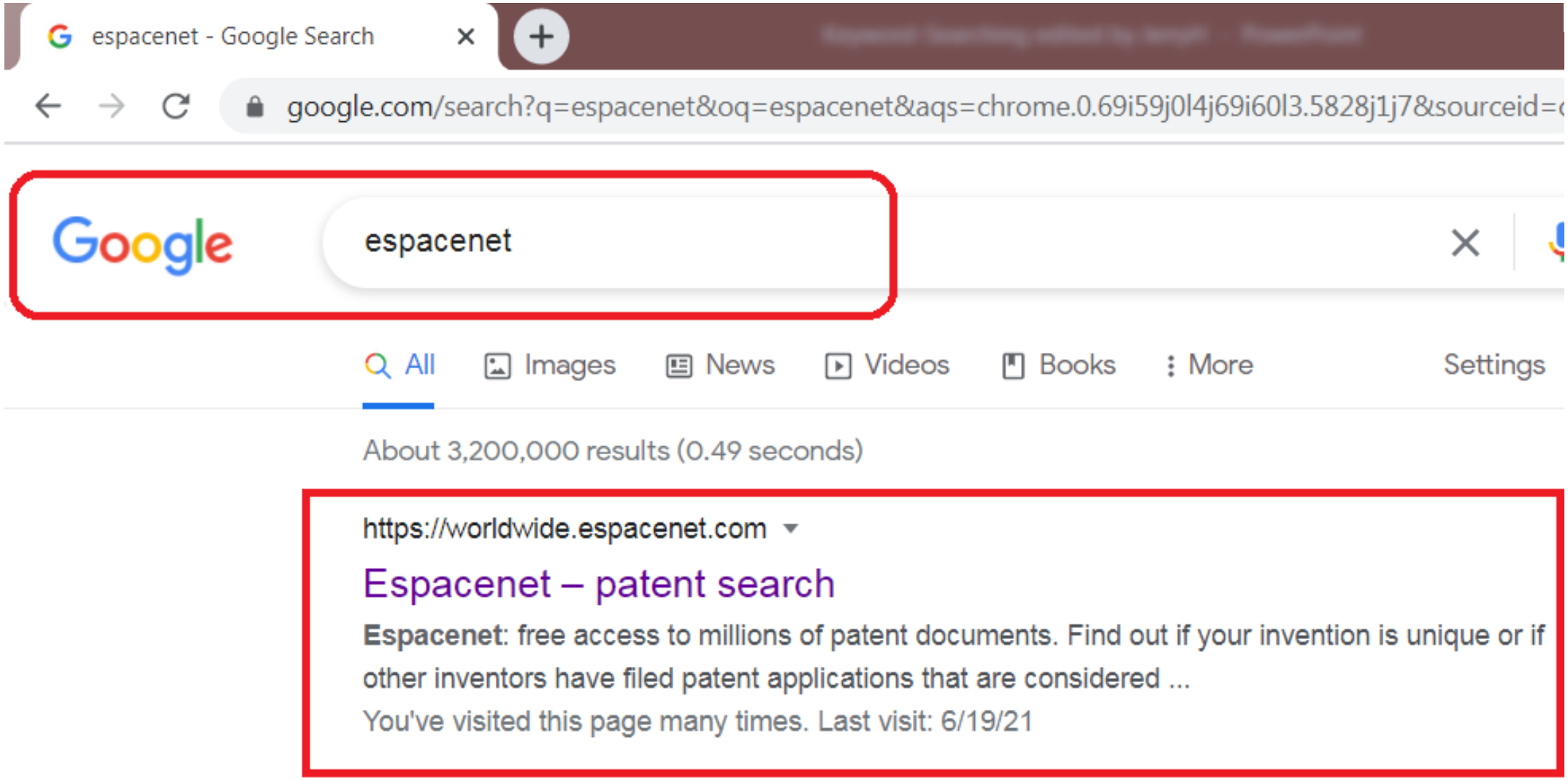
Scheme | RCL | Compilation | Catchwords | Search

(vaccin* OR inoculat* OR antigen* OR "antibody generat*")

Search | Reset

Ordered by relevance:

A61K 39/44
G01N 33/541
A61K 39/395
C07K 16/42
A61K 39/002
A61K 39/005
A61K 39/008
A61K 39/012
A61K 39/015


espacenet - Google Search
google.com/search?q=espacenet&oq=espacenet&aqs=chrome.0.69i59j0l4j69i60l3.5828j1j7&sourceid=
Google
espacenet
All
Images
News
Videos
Books
More
Settings
About 3,200,000 results (0.49 seconds)
https://worldwide.espacenet.com
Espacenet – patent search
Espacenet: free access to millions of patent documents. Find out if your invention is unique or if other inventors have filed patent applications that are considered ...
You've visited this page many times. Last visit: 6/19/21

Espacenet – patent search

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Europäisches Patentamt
European Patent Office
Office européen des brevets

Espacenet
Patent search

Enter your search terms

Office/Language

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# Espacenet: free access to over 150 million patent documents

Espacenet – search results

worldwide.espacenet.com/patent/search?q=%28nftxt%20%3D%20"vaccin%2A"%20OR%20n...

Espacenet
Patent search

(nftxt = "vaccin*" OR nftxt = "inoculat*" OR nftxt = "antigen*" OR nftxt = "

Office/Language

My Espacenet | Help | Classification search | Results | Advanced search | Filters | Popup tips | Feedback

Home > Results

**Error**

Invalid query: the maximum number of terms allowed in conjunction with field identifier "10": nftxt

# BAD VS POOR VS GOOD SEARCH

- **Bad search**
  - Title/Abstract

- **Poor search**
  - Description
  - Title/Abstract

Espacenet – search results

worldwide.espacenet.com/patent/search?q=ta%20%3D%20"mobile%20phone"

Europäisches Patentamt
European Patent Office
Office européen des brevets

**Espacenet**
Patent search

ta = "mobile phone"

My Espacenet | Help | Classification search | Results | Advanced search | Filters | Popup tips

*Home* > **Results**

**Query language: en de fr**

**AND** **+ Field**

**Title or abstract** **=**

→ **Group**

mobile phone

**Search** **Reset**

# 146 249 results found

List view: Text only | List content: All | Sort by: Relevance

*(0 patents selected)* **Select the first 20 results**

1. Mobile phone display expansion system and mobile ph...
CN204244298U • 2015-04-01 • SUN DONGMEI
**Earliest priority: 2014-12-25 • Earliest publication: 2015-04-01**
The utility model relates to a mobile phone display expansion system and a mobile phone protection cover; the mobile phone display expansion system comprises a mobile phone and the mobile phone

2. Mobile phone and display system thereof
CN104580591A • 2015-04-29 • SUN DONGMEI

Espacenet – search results

worldwide.espacenet.com/patent/search?q=desc%20%3D%20"mobile%20phone"%20AND...

**Espacenet**
Patent search

desc = "mobile phone" AND ta = "electronic device"

My Espacenet | Help | Classification search | Results | Advanced search | Filters | Popup tips

*Home* > **Results**

**Query language: en de fr**

**AND** **+ Field**

**Description** = → **Group**

mobile phone

**Title or abstract** = → **Group**

electronic device

**Search** **Reset**

## 65 626 results found

List view: Text only | List content: All | Sort by: Relevance

*(0 patents selected)* **Select the first 20 results**

1. INFORMATION PROCESSING METHOD AND ELECTRON
US2014148225A1 (B2) • 2014-05-29 • LENOVO BEIJING CO LTI
**Earliest priority: 2012-11-28 • Earliest publication: 2014-05-29**
An information processing method and an electronic device are provided. I
method, when an event to be processed is received by an electronic device
control instruction is generated. The electronic device generates prompting

2. Electronic device and information processing method
CN103716436A (B) • 2014-04-09 • LENOVO BEIJING LTD
**Earliest priority: 2014-01-13 • Earliest publication: 2014-04-09**
...The invention provides an electronic device and an information processin

# BAD VS POOR VS GOOD SEARCH

- **Good search (exhaustive)**
  - Field search is combination of title, abstract, claims and specs

Espacenet – search results

worldwide.espacenet.com/patent/search?q=nftxt%20%3D%20"mobile%20phone"%20OR%...

**Espacenet**
Patent search

nftxt = "mobile phone" OR nftxt = "cell phone" OR nftxt = "cellular phone" OR nftxt =

My Espacenet | Help | Classification search | Results | Advanced search | Filters | Popup tips

*Home* > **Results**

**Query language: en de fr**

**OR** **+ Field**

**All text fields or names** = 
→ **Group**

mobile phone

**All text fields or names** = 
→ **Group**

cell phone

**All text fields or names** = 

**Search** **Reset**

## 1 973 813 results found

List view: Text only | List content: All | Sort by: Relevance

*(0 patents selected)* **Select the first 20 results**

1. MAGNIFY HAND PHONE
KR20090036494A • 2009-04-14 • CHOI JIN TACK [KR]
**Earliest priority: 2007-10-09 • Earliest publication: 2009-04-14**
A cellular phone having a magnifying glass is provided to magnify small characters through a magnifying glass. A cellular phone has a magnifying glass and a connection unit. The connection unit connects the magnifying glass to the

2. HAND PHONE
KR20130126873A • 2013-11-21 • YOO JANG HO [KR]
**Earliest priority: 2012-05-13 • Earliest publication: 2013-11-21**
...The present invention relates to a cellular phone which comprises an earphone to be telescopically inserted into one side of the cellular phone; and a sound output mode switch to output sound using the speaker or earphone of the

3. USING METHODS AND FACILITIES OF THE MOBILE PHON

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University of Science and Technology of Southern Philippines

Technology Promotions and Commercialization Office

# KEYWORD SEARCHING

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